Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Deok-Ho Kim	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University School of Medicine			
Address	Rm. 724, Ross Research Building 720 Rutland Avenue Baltimore, MD 21205			

2. Technology Overview

Title of Proposed Technology	AI-Enabled Human Organoid and Organ-on-Chip–Based Discovery Platform for Anti-Fibrotic Therapeutics			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (V), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (V), Next-gen Communication ( ), Advanced Robotics & Manufacturing (V), Quantum ( )			
Industrial Technology Classification [1]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Bio Medical		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Pharmaceutical Bio-based Technologies and systems	R&D / Engineering Services	Testing / Inspection/ Analysis Techniques
Concept of Proposed Technology	The proposed technology is an **AI-enabled, human-relevant discovery platform** for **anti-fibrotic small-molecule drug development**, integrating **human organoid and organ-on-chip disease models** with **AI-driven virtual screening and SAR optimization**. Conventional fibrosis drug discovery pipelines rely heavily on two-dimensional cell assays and animal models, which frequently fail to recapitulate the complex cellular interactions, extracellular matrix remodeling, and biomechanical changes characteristic of human fibrotic disease. These limitations contribute to high translational failure rates in clinical development. The proposed platform establishes a **closed-loop discovery pipeline** linking: **mechanism-driven AI screening → phenotypic validation in human-relevant systems → preclinical candidate compound (PCC) nomination**. Human cardiac organoids and heart-on-chip systems are used to model key pathological features of fibrosis, including fibroblast-to-myofibroblast transition, extracellular matrix deposition, and tissue			

	stiffening. AI-based virtual screening and SAR optimization guide compound prioritization, while phenotypic readouts from organoid and organ-on-chip platforms provide translationally relevant efficacy validation. This integrated approach enables accelerated identification of anti-fibrotic compounds with improved human relevance and reduced translational risk.
Rationale for Joint R&D Project	Fibrotic diseases represent a major unmet medical need across multiple organ systems, yet effective therapeutics remain limited due to poor translation from conventional preclinical models. Advanced human organoid and organ-on-chip platforms capable of modeling fibrotic disease with high fidelity are currently available primarily at overseas research institutions and are not accessible to domestic SMEs at a practical R&D level. At the same time, Korean SMEs possess strong capabilities in **small-molecule discovery, AI-driven screening, and SAR optimization**, but lack access to globally competitive human disease validation platforms. International joint R&D is therefore essential to: • Enable access to state-of-the-art human organoid and organ-on-chip fibrosis models • Establish globally competitive preclinical validation standards • Reduce early-stage translational failure risk in anti-fibrotic drug discovery • Integrate human-relevant disease modeling with domestic AI-driven discovery pipelines Through this collaboration, the project creates a synergistic platform that accelerates PCC identification and strengthens global competitiveness in anti-fibrotic drug development.
Key Words	1) Fibrotic Diseases 2) Small-Molecule Drug Discovery 3) Human Organoid / Organ-on-Chip 4) AI-Driven Virtual Screening & SAR 5) Precision Anti-Fibrotic Therapeutics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (TRL 2)	**End of the Project**	Level (TRL 5)
R&D Objectives	1. Establish **human-relevant fibrosis disease models** using advanced organoid and organ-on-chip systems through international collaboration 2. Develop an **integrated AI-enabled small-molecule discovery pipeline** combining virtual screening, phenotypic screening, and SAR optimization 3. Identify and validate **preclinical candidate compounds (PCCs)** demonstrating robust anti-fibrotic efficacy and favorable drug-like properties 4. Create a **scalable and transferable discovery platform** applicable to multiple fibrotic diseases, including cardiac, pulmonary, and hepatic fibrosis Achievement of these objectives advances the technology to a level that enables follow-up preclinical development and global partnering by Korean SMEs.			

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R&D Target Specifications	1. Establishment of organoid and organ-on-chip fibrosis models recapitulating: • Fibroblast-to-myofibroblast transition • Extracellular matrix deposition • Increased tissue stiffness 2. Screening throughput: **≥1,000 small molecules per year** in phenotypic assays 3. Hit identification rate: **≥1–3%** with confirmed activity in human-relevant models 4. Lead optimization outcomes: **≥3–5 lead compounds** meeting all of the following criteria: • ≥50% reduction in fibrosis markers (e.g., COL1A1, ACTA2/α-SMA, FN1) • IC_{50} < 1 μM in relevant cellular assays • Acceptable preliminary ADME profiles (microsomal stability >30 minutes; no major cytotoxicity at 10 μM) • Reproducible efficacy across at least two independent human-relevant platforms (organoid + organ-on-chip) • Preliminary safety margin in cardiomyocyte viability and contractility assays
R&D Plan	**Phase 1 (Months 0–12): Platform Establishment and Initial Hit Identification** • Establish international collaboration framework and technology transfer • Standardize fibrosis induction protocols and phenotypic readouts • Perform AI-based virtual screening on compound libraries • Conduct pilot phenotypic screening to identify initial hits *Milestones:* • Functional fibrosis organoid and organ-on-chip models established **Phase 2 (Months 12–24): Hit Validation and Lead Optimization** • Secondary and tertiary screening in advanced organoid and organ-on-chip systems • Mechanism-of-action studies using transcriptomics and phenotypic profiling • SAR-driven medicinal chemistry optimization • Preliminary in vitro ADME and toxicity profiling *Milestones:* • 3–5 optimized lead compounds identified • Down-selection to top 2 lead candidates **Phase 3 (Months 24–36): PCC Nomination and Commercialization Readiness** • Cross-platform validation in independent overseas organoid and organ-on-chip systems • Ex vivo human tissue validation where available through collaborators • Final PCC nomination and preparation of preclinical development packages • Partnering strategy development and technology transfer planning *Milestones:* • ≥1 PCC nominated

	• Preclinical data package and commercialization roadmap completed

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Establish and optimize human organoid and organ-on-chip models recapitulating key fibrotic disease features • Develop standardized fibrosis induction protocols and quantitative phenotypic assays • Conduct advanced phenotypic screening and validation of candidate compounds • Provide technical know-how, protocols, and training for human-relevant preclinical evaluation • Support cross-platform validation and efficacy assessment of lead and PCC compounds
Expected R&R of Korean SME Partner	• Lead small-molecule discovery and synthesis, including compound library construction • Perform hit-to-lead medicinal chemistry and SAR optimization • Conduct AI-driven virtual screening and compound prioritization • Perform primary in vitro screening and preliminary ADME/toxicity profiling • Integrate multi-omics and phenotypic data to nominate PCCs • Lead commercialization planning and global partnering strategies

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Establishment of a globally competitive human-relevant anti-fibrotic discovery platform • Identification of novel fibrotic target pathways through multi-omics analyses • Discovery and validation of PCCs with demonstrated efficacy in human-relevant models • Generation of proprietary disease models, screening protocols, and datasets **Commercialization and Industrial Impact** • Creation of a proprietary anti-fibrotic small-molecule pipeline led by Korean SMEs • Acceleration of time-to-PCC and reduction of translational failure risk • Strengthening of Korea's drug discovery ecosystem through transferable platform technologies • Expansion of global collaboration and licensing opportunities with pharmaceutical and biotech partners **Broader Socioeconomic Impact** • Contribution to addressing high unmet medical needs in fibrotic diseases • Development of high-value R&D capabilities in AI-driven drug discovery and advanced bioengineering • Promotion of international technology co-development and global visibility in advanced bio-innovation

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6. References

(1) Travers, J. G., Kamal, F. A., Robbins, J., Yutzey, K. E., & Blaxall, B. C. (2016). Cardiac fibrosis: The fibroblast awakens. *Circulation Research*, 118(6), 1021–1040.

(2) Huh, D., Kim, H. J., Fraser, J. P., et al. (2010). Microfabrication of human organs-on-chips. *Science*, 328(5986), 1662–1668.